CSC 6740 **Data mining** **D**ue by: Sept. 11

Name:

Assignment 1 (100 points)

Q1: Compute Maximum Likelihood Estimation (MLE) for the following problem; (10pt)

If you observed I.I.D. samples $x1$, $x2$, x3, x4 , from an $Exponential(\theta)$ distribution:

PDF (x, θ) $= \theta\, exp(-\theta x) = \theta\, e^{-\theta x}$; $X > 0$

(a) What is the likelihood function?

L(x1,x2,x3,x4, θ) =

(b) What is the log likelihood function?

Ln (L(x1,x2,x3,x4, θ)) =

(c) What is formular of MLE of θ;

$\Theta =$

(d) Here is data: ($x1$,$x2$,$x3$,$x4$)=(1.0,2.5,1.5, 2.8)., plug in the data values, what is the value of θ;

Q2: *Suppose you get values x1, x2, … xn , from independent draws of a normal random variable ;* $\mathcal{N}(\mu, \theta)$ *(for* μ θ *are unknown); Please use MLS formula to derive the* μ θ *, (see slides for references) (20pt)*

Q3: You have collected a dataset with data points as the following: (please provide code and outcome plots) (35pt)

412	395	388	450	510	477	465	430	421	399
380	370	355	600	590	575	560	540	525	515
480	470	460	455	440	435	420	410	405	390
385	382	379	365	350	345	340	335	330	325
555	534	589	565	550	588	591	583	578	595
360	371	285	366	382	379	355	360	396	899

a) Plot the distribution shape using histogram with **three** different bin sizes;
b) Plot the distribution using kernel density estimation with **three** different bandwidths, one is Silverman's bandwidth, one is plug-in (SJ) bandwidth, the other of your choice.
c) show the step to compute Silverman's bandwidth manually.
d) Plot QQ plot against normal distribution;
e) Plot boxplot and What is the min, median, first quartile (Q1), third quartile, IQR ? Is there outlier based on 1.5 IQR?
f) Remove the outlier and repeat step b
g) Summarize what you have learned from these exercises

CSC 6740 **Data mining** **D**ue by: Sept. 11

Name:

Q4:. Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8): (5pt)

(a) Compute the Euclidean distance between the two objects.

(b) Compute the Manhattan distance between the two objects.

(c) Compute the Minkowski distance between the two objects, using h = 3.

(d) Compute the supremum distance between the two objects.

Q5: It is important to define or select similarity measures in data analysis. However, there is no commonly-accepted subjective similarity measure. Results can vary depending on the similarity measures used. Nonetheless, seemingly different similarity measures may be equivalent after some transformation. Suppose we have the following four-dimensional data set: (15pt)

	Att1	Att.2	Att.3	Att.4
X1	0.661	1.454	0.666	0.912
X2	0.725	1.256	0.589	1.024
X3	0.654	1.412	0.758	1.102
X4	0.632	1.896	0.729	0.987
X5	0.837	1.522	0.841	0.923

a) Consider the data as four-dimensional data points. Given a new data point, x = (0.78, 1.6, 0.7, 1.1) as a query, rank the database points based on similarity with the query using Euclidean distance, correlation, cosine similarity, and dot product. (similar data points rank first).

b) If you want to design a similarity measure based on Euclidean distance, but want this measure to have high values meaning high similarity, what you can do and why, explain in as detail as you can?

c) If you want a similarity measure to be robust to scale and translation changes, what you will do? why?

Q6: a) Explain the formula and the purpose of entropy, cross entropy and KL divergence; (15pt)

b) Explain the relation among them.

c) When is the entropy maximized for a discrete distribution?

d) Why does a cross-entropy become large when a model assigns a low probability to the true class?

e) Why is KL divergence always non-negative?